WEEK-3

Agenda:
- Expected Value
- DMC Problems
- Properties of Expection
- var and S.D
- Covariance
- Correlation
- Inequalities (Markov & Chebyshev)

Expected Value of Discrete Random Variable

$$E(x) = \sum x \cdot P(X = x)$$

→ weighted avg. of all possible values that the random variable can take.

[Long-run avg.]

Q. The probability distrib. of discrete rand. var. X is

$$f(x) = {}^4C_x \left(\tfrac{1}{4}\right)^x \left(\tfrac{3}{4}\right)^{4-x}, \quad x = 0, 1, 2, 3, 4$$

($\frac{1}{4} \to p$, $\frac{3}{4} \to q$)

Find the mean of X.

$E[x] = np$ → mean

$var[x] = np(1-p)$

Sol. $\boxed{p = \frac{1}{4}}$

mean $= np = \frac{1}{4} \cdot 4 \Rightarrow 1$

Q. # Biased die Problem

Suppose that a game is to be played with a single biased die. The no. from 2 to 6 are equally likely to land face up, but number one is thrice as likely to land face up as each of the other numbers. In the game, a player wins ₹100 if a 3 turns up, ₹150 if a 5 turns up, loses ₹160 if a 4 turns up; while the player neither wins nor loses if any other face turns up. Find the expected money to be won.

Sol.
- Numbers 2-6 (5 no.) → equal prob. $= p$
- Number 1 → Prob $= 3p$

total Prob. $= 1$ → $3p + 5p = 1$

$\boxed{p = \frac{1}{8}}$

3 → $\frac{1}{8}$

5 → $\frac{1}{8}$

4 → $\frac{1}{8}$

expected value

$$E = \frac{1}{8}(100) + \frac{1}{8}(150) + \frac{1}{8}(-160) = \frac{90}{8} = 11.25$$

Properties of Expectation

- $E[cx] = c \cdot E[x]$ for a random var. X and a const. c
- $E[X+Y] = E[X] + E[Y]$ for any 2 random var. X, Y

$E(x) \geq 0$

$E(x) = \mu$ (mean)

Q. Let X be a discrete random variable with the following probability mass fone.

$$P(X = k) = \begin{cases} 0.2 & \text{for } k = 0 \\ 0.3 & \text{for } k = 1 \\ 0.4 & \text{for } k = 2 \\ 0.1 & \text{for } k = 3 \\ 0 & \text{otherwise.} \end{cases}$$

Define $Y = (X-1)(X-2)$. Find $E[Y]$.

2
0
0
2
0

Sol. $Y = (-1)(-2) = 2$

$E(Y) = 2(0.2) + 0(\ ) + 0(\ ) + 2(0.1) + 0(0) = 0.4 + 0.2 = 0.6$

- $Var(x) = E[x^2] - E[x]^2$
  σ^2
- $s.d(x) = \sqrt{var(x)}$

Distribution	Expected value	Variance
Bernoulli (p)	p	$p(1-p)$
Binomials (n, p)	np	$np(1-p)$
Geometric (p)	$1/p$	$(1-p)/p^2$
Poisson (λ)	λ	λ
Uniform $\{1, \ldots, n\}$	$(n+1)/2$	$(n^2-1)/12$

p = success
q = failure
λ = Avg. value b/w interval

Standardized Random Variable (z-score)

A random var. x is said to standardised if $E[x] = 0$, $var(x) = 1$

$$\boxed{Y = \frac{x - E[x]}{s.d(x)}}$$

Magic Table

Operation on x	Mean	S.D/M.D	Var.
add k to each obv.	add k	unchanged	unchanged
sub. k to each obv.	sub k	unchanged	unchanged
mul. k to each obv.	mul k	mul k	mul k^2
div. k to each obv.	div k	div k	div k^2

- In x and y are Independent random var.
  and $Var(x) = var(y) = 9$
  then, $var(x-y) = var(x) + var(y) = 9+9 = 18$

Q. 3 fair coins are tossed. Find $var(x)$, where x is the no. of tails that appear.

Sol. $\{x$: No. of tails$\}$

$x^2 = \{0, 1, 4, 9\}$

$x = \{0, 1, 2, 3\}$

$P(x) = \{\frac{1}{8}, \frac{3}{8}, \frac{2}{8}, \frac{1}{8}\}$

$var = E(x^2) - [E(x)]^2 \Rightarrow 3 - (1.5)^2$
$\Rightarrow 3 - 2.25$
$\Rightarrow 0.75$

$E(x) \rightarrow$ calculate
$E(x^2) \rightarrow$ calculate

Covariance

- x and y are random variables on same probability space.

$$Cov(X,Y) = E\left[(X - E[X])(Y - E[Y])\right]$$

or $$\boxed{E(XY) - E(X)\cdot E(Y)} \star$$

- if $Cov(X,Y) \to +ve$ $\to (X - E[X])(Y - E[Y]) \to +ve$

  $X \uparrow$ $Y \uparrow$

  $X \downarrow$ $Y \downarrow$

- if $Cov(X,Y) \to -ve$ $\to (X - E[X])(Y - E[Y]) \to -ve$

  $X \uparrow$ $Y \downarrow$

  $X \downarrow$ $Y \uparrow$

- if $Cov(X,Y) = 0$

  $\hookrightarrow$ x and y are said to be "uncorrelated".

- Properties of Covariance:

  1) $Cov(X,X) = Var(X)$
  2) $Cov(X,Y) = Cov(Y,X) \to$ its symmetric
  3) $Cov(X, aY + bZ) = a\,Cov(X,Y) + b\,Cov(X,Z)$

     $Cov(aX + bY, Z) = a\,Cov(X,Z) + b\,Cov(Y,Z)$
  4) If x and y are independent, x and y are uncorrelated, $Cov(X,Y) = 0$

NV1. $\star$ if x and y are uncorrelated, they may be dependent

---

Q. Let x and y be 2 rand. var. with joint distribution given:

$y \backslash x$	0	1	2
0	$1/12$	$3/12$	0
1	$2/12$	0	$1/12$
2	$3/12$	$1/12$	$1/12$

$P(X=0) = 1/2$

$P(X=1) = 1/3$

$P(X=2) = 1/6$

$P(Y=0) = 1/3$

$P(Y=1) = 1/4$

$P(Y=2) = 5/12$

$E(XY) \Rightarrow$

$\underline{x=0} \to y=0 \Rightarrow 0 \cdot \frac{1}{12} = 0$

$1 \Rightarrow 0 \cdot \frac{2}{12} = 0$

$2 \Rightarrow 0 \cdot \frac{3}{12} = 0$

$\underline{x=1} \to y=0 \Rightarrow 1 \cdot \frac{3}{12} = 1/4$

$1 \Rightarrow 1 \cdot 0 = 0$

$2 \Rightarrow 1 \cdot \frac{1}{12} = 1/12$

$\underline{x=2} \to y=0 \Rightarrow 2 \cdot 0 = 0$

$1 \Rightarrow 2 \cdot 1/12 = 2/12$

$2 \Rightarrow 2 \cdot 1/12 = 2/12$

$E(XY) = 0 + 0 + 0 + \frac{3}{12} + 0 + \frac{1}{12} + 0 + \frac{2}{12} + \frac{2}{12} = \frac{2}{3}$

Q. The Probability distribution of a random variable x is given:

$X=x$	0	1	2	3
$P(X=x)$	1/8	1/4	1/8	1/2

Define another random variable $Y = 2X+3$. Find $Cov(X,Y)$

Sol. $Cov(X,Y) = E(XY) - E(X).E(Y)$

$E(XY) = E(X(2X+3))$

$= E(2X^2+3X)$

$= E(2X^2) + E(3X)$

$\Rightarrow 2E(x^2) + 3E(x)$

$\Rightarrow 2 \cdot \frac{21}{4} + 3 \cdot (2) = \frac{42}{4} + 6$

$\Rightarrow \frac{33}{2}$

$\boxed{E(x) = 2}$

$E(Y) = E(2x+3)$

$= 2E(x) + 3$

$= 2(2) + 3$

$\Rightarrow 7$

$\boxed{E(x^2) = \frac{21}{4}}$

$Cov(X,Y) = 2.5$

Correlation

$$\rho(X,Y) = \frac{Cov(X,Y)}{S.D(X) \cdot S.D(Y)}$$

dimensionless quantity

summarizes trend b/w 2 random var.

Range:

$-1 \le \rho(X,Y) \le 1$

$$\boxed{-SD(X) \cdot S.D(Y) \le Cov(X,Y) \le S.D(X) \cdot S.D(Y)}$$

$\rho(X,Y) \approx 0 \rightarrow$ X and Y are close to being uncorrelated

$\rho(X,Y) = 1$ or $\rho(X,Y) = -1 \rightarrow$ exists $a \neq 0$ and b so that $\boxed{Y = aX + b}$ with prob. 1

★ # Markov's Inequality $\rightarrow P(X \ge c) \le \frac{E(x)}{c}$

★ # Chebyshev Inequality $\rightarrow P(|X-\mu| \ge k\sigma) \le \frac{1}{k^2}$

or

$P(|X-\mu| \ge k) \le \frac{\sigma^2}{k^2}$

$K \rightarrow$ is a pure dimensionless no.

tells you how many S.D away from the μ

$\cancel{K = \frac{1}{\sigma}}$

Q.① A Shopkeeper sells mobile Phones. The expected demand for mobile Phones is 4 per week. Based on the given information, choose correct op.

Sol. $E(x) = 4 \approx \mu$

Markov's Inequality : $P(x \geq c) \leq \frac{\mu}{c}$

$P(x \geq 10) \leq \frac{4}{10} = 0.4$

$P(x < 10) = 1 - P(x \geq 10) = 1 - 0.4 = 0.6$

Q.② The prob. of getting heads while tossing an unfair coin is 0.40. Suppose the coin is flipped 30 times. Find an upper bound using Markov's inequality for the prob. that it lands on heads at least 20 times.

Sol. $p = 0.40$
$n = 30$

$P(x \geq 20) \leq \frac{E(x)}{20}$

$P(x \geq 20) \leq \frac{\cancel{12}\,6}{\cancel{20}\,10}$

Ans. 0.6

Q.③ Suppose a fair coin is flipped 200 times. Find an upper bound on the prob. that the no. of times the coin lands on heads is at least 130 or at most 70 using Chebyshev's inequality.

Sol. $n = 200$
$p = 0.5$
$q = 0.5$

$X \sim Bin(200, 0.5)$

$E(x) = np = 200(0.5) = 100$

$(\sigma^2)\ Var(x) = npq = 200(0.5)(0.5) = 50$

$\sqrt{\sigma^2} = 7.07$

Chebyshev inequality : $P(|x - \mu| \geq k\sigma) \leq \frac{1}{k^2}$

$P(x \geq 130)$ and $P(x < 70)$

$k = \frac{1}{6}$

$x \geq 130 \Rightarrow 130 - 100 = 30 \Rightarrow \frac{30}{6} = \frac{30}{7.07} = 4.24$

$x < 70 \Rightarrow 100 - 70 = 30 \Rightarrow 4.24$

Both should be equal

$P(|x - 100| \geq 30) \leq \frac{1}{(4.24)^2}$

easy solution: use $P(|x-\mu| \geq k) \leq \frac{\sigma^2}{k^2}$

$P(x \geq 130 \text{ or } x \leq 70)$

$\Updownarrow$

$P(|x-100| \geq 30)$

$\hookrightarrow k$

$P(|x-100| \geq 30) \leq \frac{50}{30^2} = \frac{500}{900}$

$= \frac{1}{18}$